Analytical Chemistry Cumulative Exam.
September 29, 2017
Announced Topic: Spectroscopy

Warm Up.

1. (5 points) Define isosbestic point.

2. (5 points) Define quantum yield.

3. (5 points) Define excited singlet state.

4. (5 points) Define excited triplet state.

5. (10 points) Why is signal averaging commonly applied in spectroscopic measurements? What is the chemical or spectral basis that makes this approach to data collection necessary?

6. (10 points) Use an energy diagram (Jablonski diagram) to illustrate the processes associated with absorbance, fluorescence and phosphorescence. Be sure to indicate what the relaxation processes are for each of these spectroscopic phenomena.

Problems.

7. (10 points) Provide a definition for transmittance and use an illustration if it makes your argument easier to understand.

8. (10 points) Using your definition or diagram from 7 as a guide, explain why in quantitative chemical analysis we prefer to keep solution absorbance measurements between 0 and 1.

9. (40 points) Provide a detailed explanation about how phenol red (shown below) can be used for performing an optical measurement of pH. Be sure to include chemical and/or physical information about the basis for the measurement. Be sure to comment regarding how the collected data has to be processed in order to obtain the pH value for an unknown sample. The spectra shown below are for phenol red at different pH values. The provided spectra are meant to provide an illustration and not absolute values for the measurement.

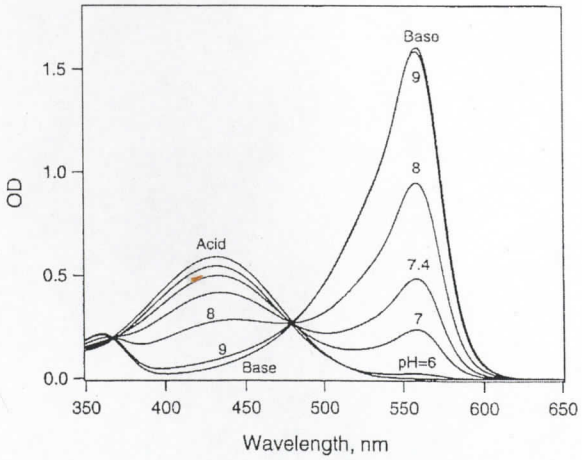
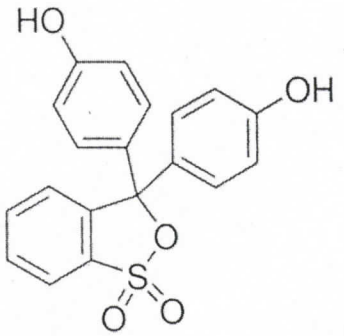


FIG. 1. Phenol red spectra in various buffer solutions at 293 °K. The acid form spectrum was obtained in a pH 4 buffer and the base form spectrum in pH 10 buffer. The dye concentration is 3.18×10^{-5} mol/kg. Spectral path length is 1 cm.



Phenol Red, $pK_a = 8.0$